

KUSHAL MITTAL

B.Tech Graduate, AI/ML Engineer & Full-Stack Developer

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EDUCATION

Poornima College of Engineering

B.Tech in Artificial Intelligence and Data Science

2022 – 2026

CGPA: 9.31 / 10.0

EXPERIENCE

CrashPlan

AI/ML Engineer Intern

March 2026 – Current

Bengaluru, India

- Built a production-grade **graph-based RAG engine** in Python/Flask for DAG-driven document ingestion and retrieval.
- Engineered a **56-node plugin framework** with parallel execution, retries, timeouts, and thread-safe model management.
- Designed **MongoDB-backed deployment infrastructure** with blue-green releases, runner caching, and execution tracking.
- Integrated **OpenAI, Elasticsearch, ChromaDB, Azure Blob, Tika, spaCy, and HuggingFace** for enterprise AI workflows.

Ernst & Young (EY)

AI/ML Intern

Jul 2025 – Jan 2026

Bengaluru, India

- Engineered scalable ML pipelines in **PySpark** on **Databricks**, cutting data preprocessing time by **30%**.
- Delivered **Generative AI solutions** with LangChain, automating client workflows and enhancing enterprise knowledge systems.
- Worked on the **Forecast Utility Engine**, implementing automated data pipelines and validation logic to streamline forecasting workflows.

National Informatics Centre (NIC)

Machine Learning Intern

May 2025 – Jul 2025

Lucknow, India

- Transformed **10K+ unstructured emails** into structured datasets using Python automation, reducing manual processing.
- Built ML classification models with **92% accuracy**, improving help-desk query categorization and resolution speed.
- Deployed scalable REST APIs to serve predictions in real-time, streamlining support workflows.

Compucom Software Limited

Backend Developer & ML Intern

Jun 2024 – Sep 2024

Jaipur, India

- Architected and tested **15+ RESTful APIs** in PHP, ensuring high system reliability and security.
- Implemented a hotel booking **cancellation prediction model** with **78% accuracy**, driving operational efficiency.
- Delivered dashboards to convert analytics into **actionable hotel insights**, improving decision-making.

PROJECTS

QA Chatbot | Python, LangChain, Google Gemini API, HuggingFace, ChromaDB, Streamlit

[GitHub](#)

- Engineered an intelligent **topic-specific assistant** using **LangChain** and **Google Gemini API**, handling **1.2K+ FAQ queries**.
- Constructed a local **vector database** with **ChromaDB** and **HuggingFace embeddings** for semantic search and efficient retrieval.
- Formulated a strict **custom RetrievalQA prompt** to prevent hallucinations, ensuring reliable and safe responses.

RAWG GameHub | React, Redux Toolkit, Clerk Auth, React-Bootstrap, RAWG API

[GitHub](#) | [Live Demo](#)

- Developed a scalable **game discovery platform** with **Clerk authentication** and RAWG API, showcasing **500+ games** dynamically.
- Implemented **real-time search**, 5+ filters, sorting, and pagination via **Redux Toolkit**, enhancing user navigation by **35%**.
- Crafted a responsive UI with **React-Bootstrap** and **custom CSS**, introducing personalized **bookmarking for 100+ games**.

Car Price Predictor | Python, Scikit-Learn, Flask, Jupyter, PyCharm

[GitHub](#)

- Constructed a machine learning pipeline predicting prices for **1,200+ cars** with Linear Regression, achieving **85% accuracy**.
- Executed data cleaning, encoding, and **feature engineering** with Pandas/NumPy, boosting model performance by **12%**.
- Launched the model via a **Flask web app**, enabling real-time price estimates from 5+ input features.

Custom CMS | PHP, MySQL, HTML, CSS, JavaScript

[GitHub](#)

- Architected a secure CMS for blog publishing with **role-based access control (admin/editor)**, managing **20+ live pages**.
- Optimized a normalized **MySQL schema** and JOIN queries for **real-time data retrieval** and fast load times.
- Executed full-stack **CRUD operations**, login authentication, and validation with PHP sessions and Bootstrap UI.

SKILLS

Programming Languages: Python, C, C++, PHP, JS, HTML, CSS, SQL

AI/ML & Data Science: LangChain, LangGraph, PySpark, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, OpenCV

Frontend & Backend: Flask, Django, React.js, Node.js, Express.js, Bootstrap

Databases: MySQL, MongoDB, PostgreSQL, Faiss, Chroma

Tools & Platforms: Git&GitHub, VS Code, PyCharm, Jupyter Notebook, Google Colab, Kaggle, Postman, Vercel, Replit

Coursework: SE, DSA, DBMS, CN, OOPS, OS, DMT, CD, AOA, CC, AI, ML, ANN

ACHIEVEMENTS AND CERTIFICATIONS

- Qualified **GATE 2025 – CS** with a score of **477** and **AIR 8217**, demonstrating strong core computer science fundamentals.
- Earned **Microsoft Certified: AI-900, AZ-900, and DP-900**, validating expertise in cloud, AI, and data concepts.
- Certified in **Prompt Design for Vertex AI**, specializing in applied Generative AI, prompt engineering, and LLM optimization.
- Solved **500+ DSA problems** on **LeetCode** and **GeeksforGeeks**, strengthening proficiency in data structures and algorithms.
- Achieved **2nd Year Batch Topper** and won **Blind Coding at Aarohan 2024** (PCE & PIET Jaipur), outperforming 60+ participants.